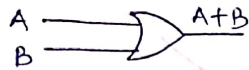


Logic gates:Basic gates:1) AND gate:

A	B	$A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

2) OR gate:

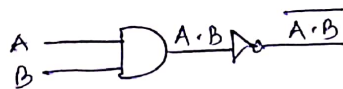
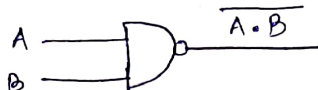
A	B	$A + B$
0	0	0
0	1	1
1	0	1
1	1	1

3) NOT gate:

A	A'
0	1
1	0

Universal gates:4) NAND gate:

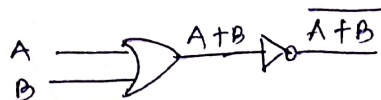
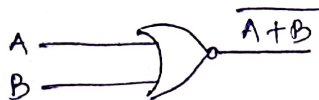
AND + NOT



A	B	$\overline{A \cdot B}$
0	0	1
0	1	1
1	0	1
1	1	0

5) NOR gate:

OR + NOT

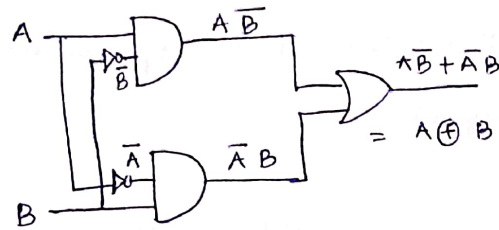


A	B	$\overline{A + B}$
0	0	1
0	1	0
1	0	0
1	1	0

Derived gates:

6) XOR gate: Exclusive OR gate

$$A \oplus B = \bar{A}B + A\bar{B}$$

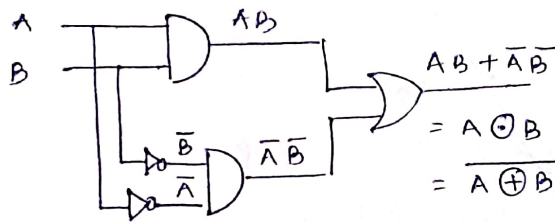


A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

7) XNOR gate: Exclusive NOR gate

Complement of XOR gate

$$A \odot B = \overline{A \oplus B} = AB + \bar{A}\bar{B}$$



A	B	$A \odot B$
0	0	1
0	1	0
1	0	0
1	1	1

Boolean laws:

1) Commutative law:
 a) $a + b = b + a$
 b) $a \cdot b = b \cdot a$

2) Associative law:
 a) $a + (b + c) = (a + b) + c$
 b) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$

3) Distributive law:
 a) $a + (b \cdot c) = (a + b) \cdot (a + c)$
 b) $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

4) Identity law:
 a) $a + 0 = a$
 b) $a \cdot 1 = a$

5) Idempotent law:
 a) $a + a = a$
 b) $a \cdot a = a$

6) Complement law:
 a) $a + a' = 1$
 b) $a \cdot a' = 0$
 c) $0' = 1$
 d) $1' = 0$

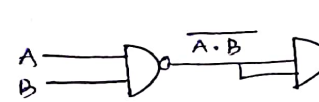
7) Boundedness law:
 a) $a + 1 = 1$
 b) $a \cdot 0 = 0$

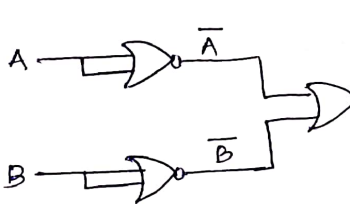
8) Absorption law:
 a) $a + a \cdot b = a$
 b) $a \cdot (a + b) = a$

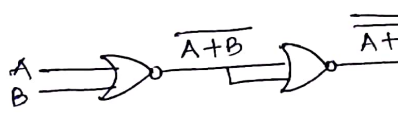
9) Involution law: $(a')' = a$

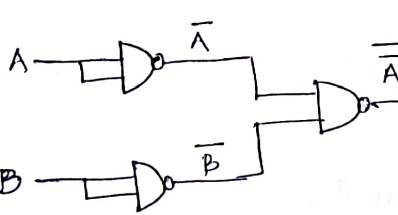
10) De Morgan's law:
 a) $(a + b)' = a' \cdot b'$
 b) $(a \cdot b)' = a' + b'$

Derivation of AND and OR gates using Universal gates:

AND using NAND:  $\overline{\overline{A \cdot B}} = A \cdot B$

AND using NOR:  $\overline{\overline{A} + \overline{B}} = \overline{\overline{A} \cdot \overline{B}} = A \cdot B$

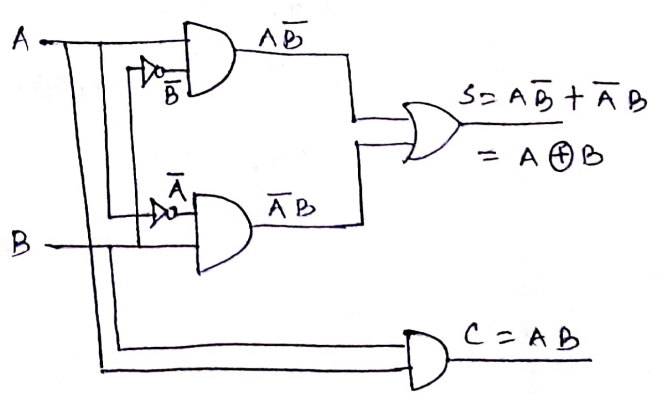
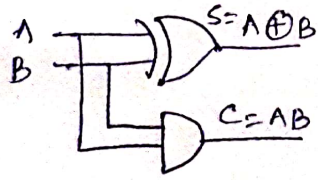
OR using NOR:  $\overline{\overline{A + B}} = A + B$

OR using NAND:  $\overline{\overline{A} \cdot \overline{B}} = \overline{\overline{A} + \overline{B}} = A + B$

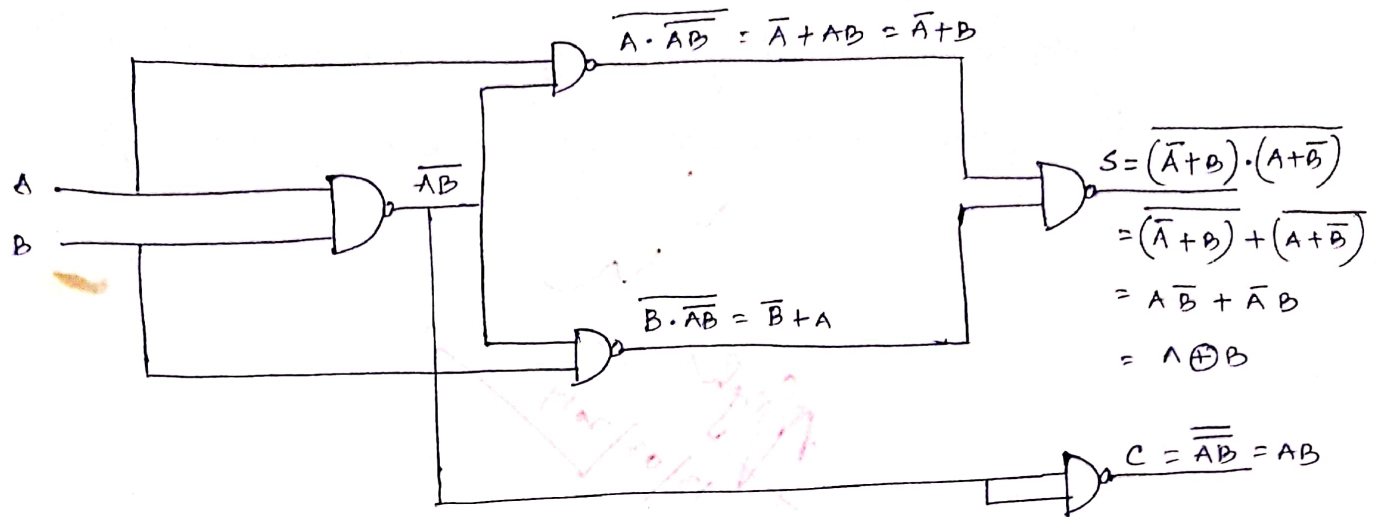
Half Adder:

		Sum Carry	
A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

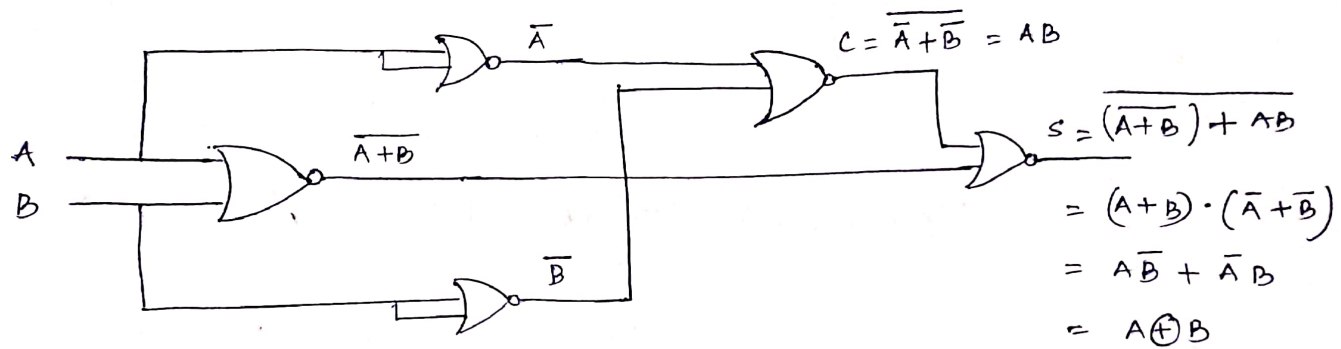
XOR AND



Half Adder using NAND gate:



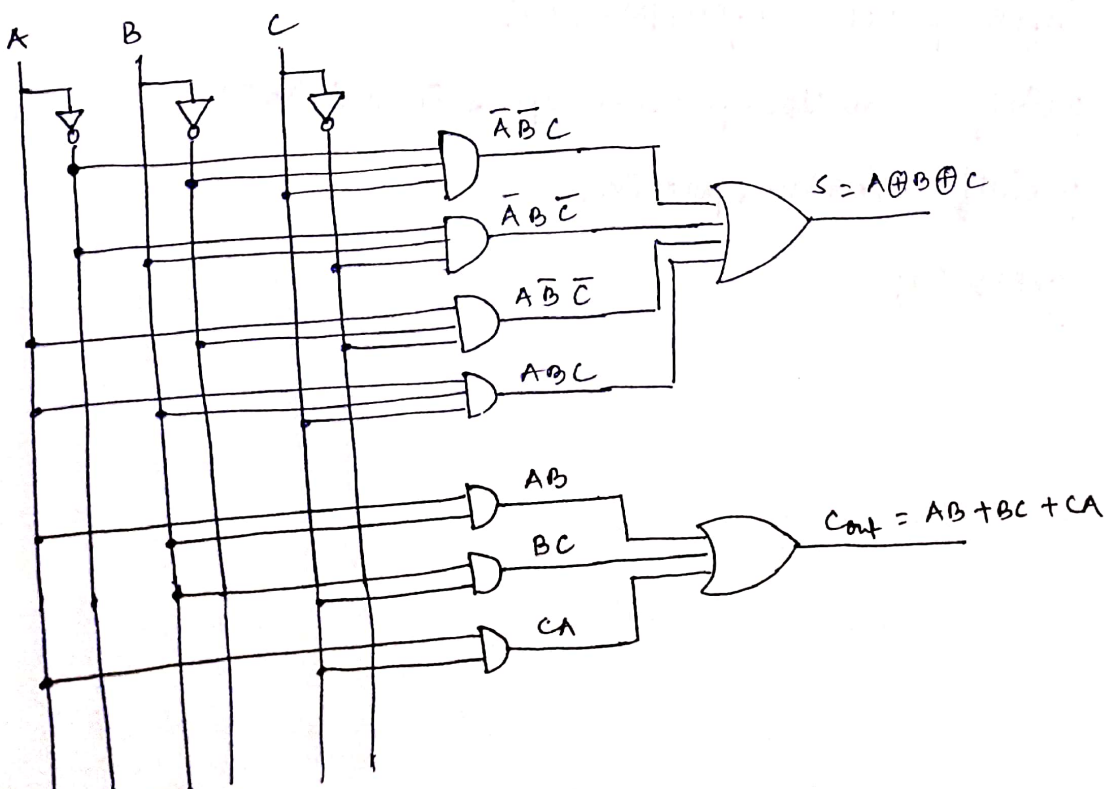
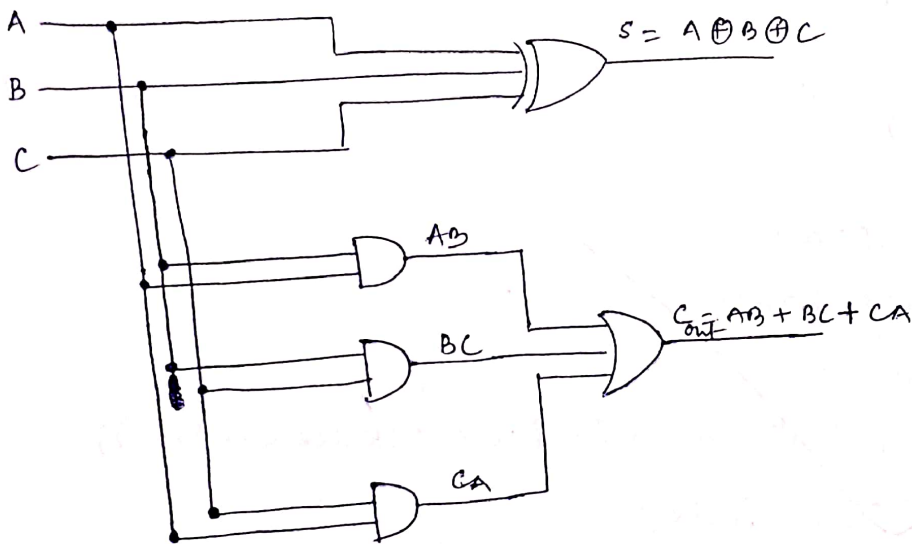
Half Adder using NOR gate:



A	B	C _{in}	S	C _{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\begin{aligned}
 S &= \bar{A}\bar{B}C + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC \\
 &= \bar{A}(\bar{B}C + B\bar{C}) + A(\bar{B}\bar{C} + BC) \\
 &= \bar{A}(B \oplus C) + A(\overline{B \oplus C}) \\
 &= A \oplus B \oplus C_{in}
 \end{aligned}$$

$$\begin{aligned}
 C_{out} &= \bar{A}BC + A\bar{B}C + AB\bar{C} + ABC \\
 &= (\bar{A}BC + ABC) + (A\bar{B}C + ABC) + (AB\bar{C} + ABC) \\
 &= BC(\bar{A} + A) + AC(\bar{B} + B) + AB(\bar{C} + C) \\
 &= AB + BC_{in} + C_{in}A
 \end{aligned}$$



(6)

proof:

$$\underline{a+a=a}$$

$$a = a+0$$

$$= a+aa'$$

$$= (a+a)(a+a')$$

$$= (a+a) \cdot 1$$

$$= a+a$$

$$\underline{a \cdot a = a}$$

$$\rightarrow a \cdot a = a \cdot a + 0$$

$$= a \cdot a + aa'$$

$$= a(a+a')$$

$$= a \cdot 1$$

$$= a$$

$$\underline{a+1=1}$$

$$\rightarrow a+1 = a+(a+a')$$

$$= (a+a)+a'$$

$$= a+a'$$

$$= 1$$

$$\underline{a+(a \cdot b) = a}$$

$$\rightarrow a+(a \cdot b) = a \cdot 1 + a \cdot b$$

$$= a \cdot (1+b)$$

$$= a \cdot 1$$

$$= a$$

$$\underline{(a+b)' = a' \cdot b'}$$

$\rightarrow$ We need to show,

$$(a+b) + a' \cdot b' = 1$$

$$\text{and } (a+b) \cdot a' \cdot b' = 0$$

$$\begin{aligned} \text{now, } (a+b) + a' \cdot b' &= [(a+b) + a'] \cdot [(a+b) + b'] \\ &= [a' + (a+b)] \cdot [a + (b+b')] \\ &= [(a' + a) + b] \cdot [a + (b+b')] \\ &= (1+b) \cdot (a+1) \\ &= 1 \cdot 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} (a+b) \cdot a' \cdot b' &= a \cdot (a' \cdot b') + b \cdot (a' \cdot b') \\ &= (a \cdot a') \cdot b' + b \cdot (b' \cdot a') \\ &= 0 \cdot b' + (b \cdot b') \cdot a' \\ &= 0 + 0 \cdot a' \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

$$\therefore (a+b)' = a' \cdot b'$$

$$\underline{(a \cdot b)' = a' + b'}$$

$\rightarrow$ We need to show,

$$(a \cdot b) + (a' + b') = 1$$

$$(a \cdot b) \cdot (a' + b') = 0$$

$$\begin{aligned} \text{now, } (a \cdot b) + (a' + b') &= [a + (a' + b')] \cdot [b + (a' + b')] \\ &= [(a + a') + b'] \cdot [b + (b' + a')] \\ &= (1 + b') \cdot [(b + b') + a'] \\ &= (1 + b') \cdot (1 + a') \\ &= 1 \cdot 1 \\ &= 1 \end{aligned}$$

$$\begin{aligned} (a \cdot b) \cdot (a' + b') &= (a \cdot b) \cdot a' + (a \cdot b) \cdot b' \\ &= (b \cdot a) \cdot a' + a \cdot (b \cdot b') \\ &= b \cdot (a \cdot a') + a \cdot (b \cdot b') \\ &= b \cdot 0 + a \cdot 0 \\ &= 0 + 0 \\ &= 0 \end{aligned}$$

$$\therefore (a \cdot b)' = a' + b'$$



Sum of Products (SOP):

- A Boolean Expression E is said to be in a sum of products form if E is a sum of two or more products of variables (complemented or uncomplemented)

Ex: $xz' + x'y'z' + xy'z$

Product of Sums (POS):

- A Boolean Expression E is said to be in a product of sums ^{form} if it consists of several sum terms logically multiplied. The variables may or may not be complemented.

Ex: $(x+y)(x+z)(x+y')$

Normal Form:

Minterm: A minterm of n variables is a product of n literals in which each variable appears exactly once in either true or complemented form, but not both.

For ex, for two variables x and y , minterms are $xy, x'y, xy', x'y'$

- In a similar way, n variables can be combined to form 2^n minterms.

Maxterm: A maxterm of n variables is a sum of n literals in which each variable appears exactly once in either true or complemented form, but not both.

For ex, for two variables x and y , maxterms are

$$x+y, x'+y, x+y', x'+y'$$

- Here also for n variables, 2^n maxterms can be formed.

- When a Boolean function f is written as a sum of minterms, it is referred to as the disjunctive normal form (DNF) of the Boolean function. It is also called canonical sum of products or standard or products.

(8)
- When a Boolean function f is written as a product of maxterm, it is referred to as the conjunctive normal form (CNF) of the Boolean function. It is also called canonical product of sums and standard product of sums.

Expression of a Boolean function as a canonical form:

From a given truth table:

Ex: Consider the truth table given and find out the corresponding Boolean expression in sum of minterms and product of maxterms.

X	Y	Z	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

→ Sum of minterms: Corresponding to 1 entries of F

$$F = x'y'z' + x'y'z + x'yz' + xy'z' + xy'z$$

Product of maxterms: Corresponding to 0 entries of F

$$F = (x + y' + z') \cdot (x' + y' + z) \cdot (x' + y + z')$$

Algebraic Method:

- To obtain the minterm expression by algebraic method, first write the expression as a sum of products and then introduce the missing variable in each term by applying the theorem $a + a' = 1$.

(9)

- For finding maxterm expansion, factor the function to obtain a product of sums, introduce the missing variable in each sum term by using $aa' = 0$ and then factor again to obtain the maxterms.

Ex: Express the Boolean function $f(x, y, z) = x + y'z$ in a sum of minterms.

→ The function has three variables x, y and z .

$$\text{Then } f(x, y, z) = x + y'z$$

$$= x(y + y')(z + z') + y'z(x + x')$$

$$= (xy + xy')(z + z') + xy'z + x'y'z$$

$$= xyz + xyz' + xy'z + xy'z' + x'y'z + x'y'z$$

$$= xyz + xyz' + xy'z + xy'z' + x'y'z$$

Ex: Express the Boolean function $f(a, b, c) = ab + a'c$ as a product of maxterms.

→ $f(a, b, c) = ab + a'c$

$$= (ab + a')(ab + c)$$

$$= (a + a')(b + a') + (a + c)(b + c)$$

$$= (b + a')(a + c)(b + c)$$

$$= (b + a' + cc')(a + c + bb')(b + c + aa')$$

$$= (b + a' + c)(b + a' + c')(a + c + b)(a + c + b')(b + c + a)(b + c + a')$$

$$= (a' + b + c)(a' + b + c')(a + b + c)(a + b' + c)$$

Conversion of DNF to CNF:

- 1) Denote the given function which is in DNF by f .
- 2) Find f' , which is the sum of those terms of the complete DNF in the variables of f , that are not present in the given function.
- 3) Take complement f'' of f' .
- 4) Simplify R.H.S. of f'' by using De Morgan's laws. This will transform the R.H.S. into its CNF.
- 5) Since $f'' = f$, the R.H.S. is the required CNF of the given function in DNF.

Conversion of CNF to DNF:

- 1) Denote the given function which is in CNF by f .
- 2) Find f' , which is the product of those terms of the complete CNF in the variables of f , that are not present in the given function.
- 3) Take complement f'' of f' .
- 4) Simplify the R.H.S. of f'' by using De Morgan's laws. This will transform the R.H.S. into its DNF.
- 5) Since $f'' = f$, the R.H.S. is the required DNF of the given function in CNF.

Ex: Convert the Boolean function $f(x, y) = xy' + x'y + x'y'$ to its conjunctive normal form.

→ Here, $f(x, y) = xy' + x'y + x'y'$

Hence, $f'(x, y) = xy$

So, $f''(x, y) = (xy)' = x' + y'$ which is the required CNF

Ex: Convert the Boolean function

$$f(x, y, z) = (x' + y + z')(x' + y + z)(x + y' + z) \text{ in disjunctive normal form.}$$

→ Here $f(x, y, z) = (x' + y + z')(x' + y + z)(x + y' + z)$

Hence, $f'(x, y, z) = (x + y + z)(x + y + z')(x + y' + z')(x' + y' + z)(x' + y' + z')$

$$\therefore f''(x, y, z) = [(x + y + z)(x + y + z')(x + y' + z')(x' + y' + z)(x' + y' + z')]'$$

$$= [(x + y + z)(x + y + z')(x + y' + z')(x' + y' + z)]' + xyz$$

$$= [(x + y + z)(x + y + z')(x + y' + z')] + xyz' + xyz$$

$$= [(x + y + z)(x + y + z')] + x'y'z + xyz' + xyz$$

$$= (x + y + z)' + x'y'z + x'y'z + xyz' + xyz$$

$$= x'y'z' + x'y'z + x'y'z + xyz' + xyz$$

which is the required DNF.

Karnaugh Map (K-Map)

- K-Map is a graphical technique which provides procedure for simplification of Boolean expressions of two, three or four variables.
- It is a diagram made up of a number of squares.
- If the expression contains 'n' variables, the map will have 2^n squares.
- Each square represents a minterm.
- 1's are written in the corresponding squares for the minterms present in the expression and 0's are written in those squares which correspond to the minterms not present in the expression.

2 - variable K-Map:

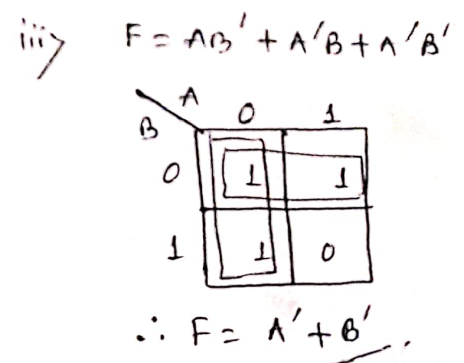
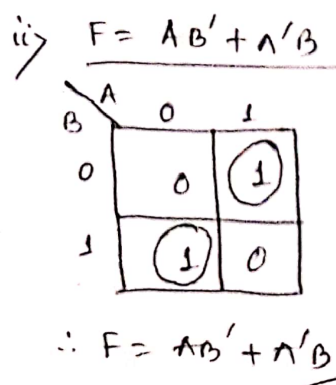
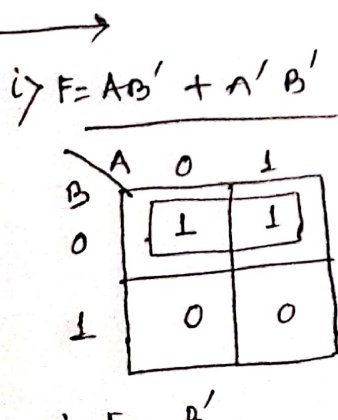
- Since the number of variables are 2, the map will have $2^2 = 4$ squares.

Ex: Find K-Maps and simplify the expressions

i) $AB' + A'B'$

ii) $AB' + A'B$

iii) $AB' + A'B + A'B'$



3- Variable K-Map:

(13)

- Since the no. of variables are 3, the map will have $2^3 = 8$ squares.
- While forming groups of squares containing 1s the following considerations must be kept
 - i) The no. of squares in a group must be equal to 2^n such that 2, 4, 8, 16.
 - ii) A square containing 1 can be included in as many groups as desired.
 - iii) Groups must be largest possible groups, a group of two squares containing 1 should not be made if these squares can be included in a group of four squares. (quad)

Ex.: Use K-Map to simplify the followings:

a) $X = ABC' + ABC$

b) $X = A'B'C' + AB'C'$

c) $X = A'B'C' + A'BC' + ABC' + AB'C$

d) $X = A'B'C' + A'B'C + A'BC + A'BC' + AB'C + ABC$

a) $X = ABC' + ABC$

AB \ C	00	01	11	10
0	0	0	1	0
1	0	0	1	0

$\therefore X = AB$

b) $X = A'B'C' + AB'C'$

AB \ C	00	01	11	10
0	1	0	0	1
1	0	0	0	0

$\therefore X = B'C'$

c) $X = A'B'C' + A'BC' + ABC' + AB'C$

AB \ C	00	01	11	10
0	1	1	1	0
1	0	0	0	1

$\therefore X = A'C' + BC' + AB'C$

d) $X = A'B'C' + A'B'C + A'BC + A'BC' + AB'C + ABC$

AB \ C	00	01	11	10
0	1	1	0	0
1	1	1	1	1

$\therefore X = A' + C$

4 - variable K-Map :

- Since the no. of variables are 4, the map will have $2^4 = 16$ squares. (octet)

Ex.: Use K-map to simplify the following expression:

15

a) $X = A'B'CD + A'B'CD' + AB'C'D' + AB'CD'$

b) $X = A'BC'D' + ABC'D' + A'BCD' + ABCD'$

c) $X = A'B'C'D' + AB'C'D' + A'B'CD' + AB'CD'$

a) $X = A'B'CD + A'B'CD' + AB'C'D' + AB'CD'$

AB \ CD	00	01	11	10
00				1
01				
11	1			
10	1			1

$\therefore X = A'B'C + AB'D'$

b) $X = A'BC'D' + ABC'D' + A'BCD' + ABCD'$

AB \ CD	00	01	11	10
00		1	1	
01				
11				
10		1	1	

$\therefore X = BD'$

c) $X = A'B'C'D' + AB'C'D' + A'B'CD' + AB'CD'$

(16)

AB \ CD	00	01	11	10
00	1			1
01				
11				
10	1			1

$\therefore X = B'D'$

Ex.: Use K-Map to simplify the Boolean expression

$$f = a'b'c'd' + ab'c'd' + a'bc'd' + a'b'cd + a'bcd + abcd + ab'cd + a'b'cd' + a'bcd' + abcd' + ab'cd'$$

ab \ cd	00	01	11	10
00	1			1
01		1		
11	1	1	1	1
10	1	1	1	1

$\therefore f = c + b'd' + a'bd$

Ex.: Simplify the Boolean function

$$F(A, B, C, D) = \sum(0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11)$$

$F = A' + B'C' + B'D$

Ex.: Draw K-map and simplify the following Boolean expressions: (17)

a) $F(A, B, C) = \Sigma (2, 3, 6, 7)$

[Ans: B]

b) $F(A, B, C) = \Sigma (0, 1, 3, 4, 5)$

[Ans: $B' + A'C$]

c) $F(A, B, C, D) = \Sigma (7, 13, 14, 15)$

[Ans: $ABD + ABC + BCD$]

d) $F(A, B, C, D) = \Sigma (0, 2, 4, 6, 8, 10, 12, 14, 15)$

[Ans: $D' + ABC$]

— X —

Types of Boolean f/c

POS form (Prod of sum form)

$$(\bar{x}_1 + x_2 + x_3) \cdot (\bar{x}_1 + \bar{x}_2 + x_3) \cdot (x_1 + x_2 + x_3)$$

or

$$(\bar{x}_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee \bar{x}_2 \vee x_3) \wedge (x_1 \vee x_2 \vee x_3)$$

max term

conjunctive Normal form (CNF)

A boolean expression is said to be CNF if it is meet of maxterms. or (product of variables & their negations)
or (product of elementary sum)

SOP form (sum of product form)

$$(x_1 \cdot x_2 \cdot x_3) + (\bar{x}_1 \cdot \bar{x}_2 \cdot x_3) + (x_1 \cdot x_2 \cdot x_3)$$

or

$$(\bar{x}_1 \wedge x_2 \wedge x_3) \vee (\bar{x}_1 \wedge \bar{x}_2 \wedge x_3) \vee (x_1 \wedge x_2 \wedge x_3)$$

min term

Disjunctive Normal form (DNF)

A Boolean expression is said to be DNF if it is join of min terms. or (sum of variables & their negation)
(sum of elementary products)

Note:-

$$P \rightarrow Q \equiv P' \vee Q$$

$$P \leftrightarrow Q \equiv P \rightarrow Q \wedge Q \rightarrow P \quad \text{is CNF}$$

$$\equiv (P \wedge Q) \vee (P' \wedge Q') \quad \text{is DNF}$$

CNF $\Rightarrow$ DNF

eg, CNF, $f(x, y, z) = (x+y+z)(x+y+z')(x+y'+z)$ convert this to DNF.

(sol)

$$(x+y+z)(x+y+z')(x+y'+z)(x+y+z')(x+y'+z')$$

$$f'(x, y, z) = (x+y+z)(x+y+z')(x'+y'+z) \\ (x+y'+z')(x'+y'+z)$$

$$f''(x, y, z) = [(x+y+z) \cdot (x+y+z') \cdot (x'+y'+z) \cdot (x'+y'+z')]$$

$$= (x+y+z)' + (x+y+z')' + (x'+y'+z)' \\ + (x'+y'+z')'$$

$$= x'y'z' + x'y'z + x \cdot y \cdot z' + x \cdot y \cdot z + x'yz$$

$$\equiv \text{DNF}$$